Bo Yang

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EDUCATION

Purdue University, College of Pharmacy 08/2023-Present
Ph.D. in Medicinal Chemistry & Molecular Pharmacology

University of Pittsburgh, School of Pharmacy 08/2021-05/2023
M.S. in Pharmaceutical Science

East China University of Science and Technology 09/2017-06/2021
B.S. in Pharmaceutical Engineering

SKILLS

Machine/Deep Learning: Graph neural networks, large language models (LLM), flow matching models, diffusion models, VAEs; reinforcement learning (GRPO/PPO), LoRA; agentic LLM systems with tool-use orchestration.
Programming & cheminformatics: Python, PyTorch, PyTorch Geometric, HuggingFace Transformers, Unsloth, RDKit; SLURM, distributed multi-GPU training, large-scale dataset curation; Unix/Linux, R.
Molecular Modeling: GROMACS, AMBER, Schrödinger Maestro, AutoDock Vina, MOE; ESM-2, Boltz2, Odesign.
Experimental Skills: Organic synthesis, synthetic route design, IR, UV, HPLC, LC/MS, NMR.
Language: English (fluent), Mandarin (native).

RESEARCH EXPERIENCE

Graduate Research Assistant, Dr. Jianing Li Lab, Purdue University 01/2024-Present

- **Benchmarking 3D generative models**: Established benchmarking metrics and evaluation frameworks for structure-based molecular generative algorithms, effectively identifying chemical plausibility limitations of existing SOTA methods and proposing strategies for enhanced performance.
- **Slogen:** Unified fragment generation and screening under structure-based scheme (1.8 M molecule-protein graph pairs) with a graph neural network, showing that generation and screening together complement human-expert design in exploring chemical space; designs from Slogen were confirmed to bind the target with high affinity while opening novel SAR space.
- **StructureSAFE:** Brought SAFE representation and pocket conditioning to a chemical language model pretrained on 1.5 B molecules and fine-tuned on 0.7 M protein–ligand complexes, first time enabling structure-conditioned molecular generation and optimization on sequence representations within a single framework, with improved chemical plausibility and drug-like properties over existing methods. Now applied across multiple projects to propose molecules for synthesis and test.
- **AlloGen:** Introduced electrostatic potential (ESP) as a physicochemically meaningful latent representation for 3D structure-based generation via flow matching, enabling design of GPCR lipid-facing allosteric modulators from limited data with a more controllable generation process.
- **Chemfixer (chemistry-aware inference-time steering):** Built an inference-time steering framework for a 3D molecular generator that improves the chemical plausibility of generated structures while enabling exploration of novel scaffolds, injecting medicinal chemistry priors into the generation process.
- **Species evolution and natural products:** Embedded natural products (NPs) and species evolutionary information into a shared latent space with a heterogeneous graph neural network, enabling a chemical view of evolutionary relationships and helping identify previously unknown evolutionary events.
- **PAC1R allostery:** Ran large-scale all-atom GROMACS simulations for PAC1R in explicit membranes, identifying previously unreported cholesterol-binding pockets with implications for class B GPCR allosteric regulation.
- **Additional work**: Fine-tuned an LLM with RL (Qwen3 + LoRA) for protein-conditioned neuropeptide generation, and co-developed a lab LLM-orchestrated observe–think–act–evaluate agent that autonomously chains tools for binder design.

Molecular Design Intern, GSK, Upper Providence 05/2026-08/2026

- **Oligonucleotide conformational sampling:** Developed and deployed a first-of-its-kind automated sampling pipeline for

highly modified oligonucleotides, replacing a manual process and delivering high-quality 3D structures at scale for binding energy estimation and structure-based design.

- **Binding energy benchmarking:** Evaluated and cross-compared binding energy estimation methods (MM/GBSA, shape complementarity, electrostatic complementarity), correlating computed scores with experimental assay endpoints to identify which metrics were predictive enough to guide synthesis prioritization.

Graduate Research Assistant, Dr. Jaden Jun lab, University of Pittsburgh 09/2021-04/2023

- **CB2R allosteric modulators (AM) design and synthesis:** Designed and synthesized >50 compounds as potential CB2R allosteric modulators, applying synthetic route design and multi-step organic synthesis in collaboration with pharmacologists.
- **Structure-based AM design:** Employed molecular docking, MD simulations, and SAR analysis to rationalize binding mechanisms and optimize pharmacological profiles, improving data-driven decision making in hit-to-lead optimization.

SELECTED PUBLICATIONS

- **Yang, B.** et al. Structure-Based Generation of 3D Small-Molecule Drugs: Are We There Yet? Journal of Medicinal Chemistry 2025, 68 (21), 22756-22768.
- Tao, H.; **Yang, B.** et al. Covalent-Allosteric Inhibitors: Do We Get the Best of Both Worlds? Journal of Medicinal Chemistry 2025, 68 (4), 4040-4052. (Co-first Author, Supplementary Cover)
- **Yang, B**. et al. StructureSAFE: A structure-aware chemical language model for unified hit identification and lead optimization. bioRxiv 2026, 2026.2006.2028.735128.
- **Yang, B**. et al. SLOGEN: A Structure-based Lead Optimization Model Unifying Fragment Generation and Screening. bioRxiv 2025, 2025.10.14.682343.
- **Yang, B.** et al. Evolutionary Tree in Chemical Space of Natural Products. bioRxiv 2025, 2025.2008.2031.673394.
- Hamilton, N. B.; **Yang, B.** et al. Molecular Basis of PAC1R Allosteric Modulation with Lipids in Membranes. bioRxiv, 2025, 2025.2001.2027.635027.
- Zhu, Y. ... **Yang, B.** et al. Backmapping of Highly Coarse-Grained Protein Models and Applications to Nanomedicine Designs. Journal of Chemical Theory and Computation 2026, 22 (15), 7940-7954.
- Yuan, J.; **Yang, B.** et al. Targeting the endocannabinoid system: Structural determinants and molecular mechanism of allosteric modulation. Drug Discovery Today 2023, 28 (7), 103615.
- **Yang, B.** et al. System and methods for generation of fragment module libraries for lead optimization of pharmaceutically active molecules. US Patent US 2026/0080976 A1.